

AP Vidyut Recruitment Exam 23rd to 27th Aug 2026

Hall Ticket Number	APVR0048215
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Test Center Name	KSRM College of Engineering
Test Time	9:00 AM - 12:00 PM
Test Date	27/08/2026
Subject	Electrical Engineering

Section : General Intelligence and Reasoning

Q.1 Which of the following letter-number clusters will replace the question mark (?) in the given series to make it logically complete?

AWE97, JFN105, SOW113, BXF121, ?

- Ans
- ☒ 1. KHO129
 - ☒ 2. KGO129
 - ☒ 3. EGO129
 - ☒ 4. KGP129

Question ID : 441009324256
Chosen Option : 2

Q.2 A, B, C, D, E, F and G are sitting around a circular table facing the centre. Only two people sit between A and B when counted from the right of A. Only two people sit between B and E. Only three people sit between A and G. F sits to the immediate left of C. How many people sit between E and F when counted from the left of F?

- Ans
- ☒ 1. Two
 - ☒ 2. One
 - ☒ 3. Three
 - ☒ 4. Four

Question ID : 441009774085
Chosen Option : 2

Q.3 Select the number from among the given options that can replace the question mark (?) in the following series.
13, 18, 28, 43, 63, ?

- Ans
- ☒ 1. 86
 - ☒ 2. 88
 - ☒ 3. 90
 - ☒ 4. 84

Question ID : 441009140864
Chosen Option : 2

Q.4 Refer to the following number, symbol series and answer the question. All numbers are single-digit numbers.

(Left) % 8 4 # @ 5 & 9 3 Σ 2 Ω 1 7 * \$ 6 ^ (Right)

If all the symbols are dropped from the series, which of the following will be sixth from the right?

- Ans
- ☒ 1. 3
 - ☒ 2. 5
 - ☒ 3. 2
 - ☒ 4. 9

Question ID : 441009796131
Chosen Option : 4

Q.5 Which of the following letter-number clusters will replace the question mark (?) in the given series to make it logically complete?

SXC53, BGL44, KPU35, TYD26, ?

- Ans
- ☒ 1. CHM17
 - ☒ 2. CIM17
 - ☒ 3. CHM16
 - ☒ 4. DHM17

Question ID : 441009622
Chosen Option : 1

Section : General awareness

Q.6 Who was elected to the British House of Commons in 1892 and played a key role in presenting the Indian viewpoint through the British Committee of the Indian National Congress?

- Ans
- ☒ 1. G.K. Gokhale
 - ☒ 2. Dadabhai Naoroji
 - ☒ 3. Subramania Iyer
 - ☒ 4. Mahadev Govind Ranade

Question ID : 44100968588
Chosen Option : 2

Q.7 What is the theme for National Technology Day 2026?

- Ans
- ☒ 1. Science and Technology for Sustainable Future
 - ☒ 2. Responsible Innovation for Inclusive Growth
 - ☒ 3. Digital India for a New Era
 - ☒ 4. Integrated Approach in S&T for Sustainable Future

Question ID : 441009310611
Chosen Option : 4

Q.8 Which type of unemployment is commonly found in rural agricultural areas in India?

- Ans ☒ 1. Seasonal Unemployment
☐ 2. Structural Unemployment
☐ 3. Cyclical Unemployment
☐ 4. Frictional Unemployment

Question ID : 44100969978
Chosen Option : 1

Q.9 The Charter Act of 1833 is notable because of which of the following?

- Ans ☐ 1. Granted dominion status to British Indian territories
☐ 2. Established provincial autonomy in major presidencies
☐ 3. Introduced separate electorates for religious communities
☒ 4. Abolished the East India Company's trade monopoly

Question ID : 4410092115938
Chosen Option : 3

Q.10 Which one of the following is NOT a biosphere reserve of India recognised by UNESCO's Man and the Biosphere (MAB) programme as of July, 2026?

- Ans ☐ 1. Nanda Devi Biosphere Reserve
☒ 2. Jim Corbett Biosphere Reserve
☐ 3. Gulf of Mannar Biosphere Reserve
☐ 4. Simlipal Biosphere Reserve

Question ID : 4410092115944
Chosen Option : 4

Q.11 In which state is the famous Pattachitra painting tradition primarily practiced?

- Ans ☐ 1. Assam
☐ 2. Kerala
☒ 3. Odisha
☐ 4. Nagaland

Question ID : 44100980722
Chosen Option : 3

Q.12 Which climatic factor and plant adaptation best explain the distribution of tropical deciduous forests in India?

- Ans ☒ 1. Seasonal variation in rainfall determines leaf shedding and regeneration
☐ 2. Soil salinity controls species adaptation in the region
☐ 3. Proximity to rivers affects forest density and diversity
☐ 4. Altitude influences plant composition and distribution

Question ID : 4410092258042
Chosen Option : 4

Comprehension:

Read the passage carefully and answer the questions that follow.

The rapid proliferation of AI-generated deepfakes has transformed misinformation from an issue of content verification into one of institutional trust. While advances in artificial intelligence have expanded opportunities for creativity and communication, they have simultaneously enabled highly convincing fabrications capable of influencing elections, financial markets, and public opinion. Consequently, governments are increasingly complementing legal sanctions with technological authentication systems, digital literacy initiatives, and international cooperation. Yet experts argue that no regulatory framework can remain effective unless citizens themselves develop the capacity to evaluate digital content critically. Moreover, independent journalism, transparent platform governance, and continuous public awareness campaigns are becoming indispensable safeguards against sophisticated digital manipulation. The challenge, therefore, extends beyond detecting manipulated media to cultivating resilient democratic institutions capable of sustaining public confidence despite increasingly sophisticated forms of digital deception.

SubQuestion No : 13

Q.13 Which of the following best expresses the central theme of the passage?

- Ans** ✓ 1. Combating AI-generated misinformation requires strengthening institutional resilience and public critical literacy alongside technological and legal countermeasures.
- ✗ 2. The principal consequence of deepfake technology lies in disrupting financial markets rather than undermining democratic institutions and public confidence generally.
- ✗ 3. International cooperation has become the primary solution to digital deception since domestic institutions lack meaningful authority over emerging artificial-intelligence technologies.
- ✗ 4. Artificial intelligence has made digital misinformation inevitable because technological innovation consistently advances more rapidly than governmental regulatory interventions worldwide.

Question ID : 4410092413879
Chosen Option : 1

Comprehension:

Read the passage carefully and answer the questions that follow.

The rapid proliferation of AI-generated deepfakes has transformed misinformation from an issue of content verification into one of institutional trust. While advances in artificial intelligence have expanded opportunities for creativity and communication, they have simultaneously enabled highly convincing fabrications capable of influencing elections, financial markets, and public opinion. Consequently, governments are increasingly complementing legal sanctions with technological authentication systems, digital literacy initiatives, and international cooperation. Yet experts argue that no regulatory framework can remain effective unless citizens themselves develop the capacity to evaluate digital content critically. Moreover, independent journalism, transparent platform governance, and continuous public awareness campaigns are becoming indispensable safeguards against sophisticated digital manipulation. The challenge, therefore, extends beyond detecting manipulated media to cultivating resilient democratic institutions capable of sustaining public confidence despite increasingly sophisticated forms of digital deception.

SubQuestion No : 14

Q.14 Which of the following situations best reflects the principle advanced in the passage?

- Ans** ✗ 1. A country permanently prohibits artificial-intelligence research, expecting technological restrictions alone to eliminate fabricated digital content across online platforms.
- ✗ 2. Public institutions rely exclusively upon criminal penalties against content creators while disregarding digital-literacy programmes and technological verification mechanisms completely.
- ✗ 3. Social-media companies remove every controversial post automatically, assuming that extensive censorship inevitably strengthens democratic trust among citizens over time.
- ✓ 4. A government introduces verified digital-content authentication, media-literacy education, and coordinated regulations to reduce misinformation during national elections.

Question ID : 4410092413880
Chosen Option : 4

Q.15 Complete the following using suitable option for the direct speech.

He said, "Mother, can I please have a chocolate?"

He_

- Ans
- ☐ 1. requested his mother asking her if he could have a chocolate.
 - ☐ 2. asked his mother requesting her if he could have a chocolate.
 - ☐ 3. requested mother asking her if he could have a chocolate.
 - ☒ 4. requested his mother for a chocolate.

Question ID : 4410091690213
Chosen Option : 2

Q.16 Complete the following sentence by choosing the correct option.

In academic writing, it is important to avoid ____ language that could mislead readers.

- Ans
- ☐ 1. explicit
 - ☐ 2. coherent
 - ☒ 3. ambiguous
 - ☐ 4. concise

Question ID : 4410091711093
Chosen Option : 4

Q.17 Substitute one word for the given underlined segment by choosing from the following options.

International organizations often distribute decision-making authority away from a central government to regional or local bodies to increase administrative efficiency.

- Ans
- ☐ 1. consolidate
 - ☒ 2. devolve
 - ☐ 3. ratify
 - ☐ 4. annex

Question ID : 4410091713148
Chosen Option : 4

Section : Quantitative aptitude

Q.18 On a circular track of length 2340 m, Raman completes one round in 45 seconds and Mohan completes one round in 130 seconds. If they start together and run in the same direction, how many different meeting points will occur on the circumference?

- Ans
- ☐ 1. 26
 - ☒ 2. 17
 - ☐ 3. 14
 - ☐ 4. 20

Question ID : 441009538574
Chosen Option : 4

Q.19 A water tank built by a municipality to supply water to 28,000 inhabitants at the rate of 125 litres per person per day is 48 m long and 35 m broad. When the tank is full, it can supply water for 4 days to the inhabitants of the town. Find the depth of the tank. (Round off to two decimal places.)

- Ans
- ☒ 1. 6.22 m
 - ☒ 2. 8.33 m
 - ☒ 3. 7.34 m
 - ☒ 4. 9.56 m

Question ID : 4410091556680

Chosen Option : 2

Q.20 Three friends P, Q and R started a business by investing a sum of money in the ratio of 2 : 4 : 7. After 6 months Q withdraws half of his capital. If the sum invested by P is ₹32000, out of a total annual profit of ₹78000 find the share of Q in profit.

- Ans
- ☒ 1. ₹20935
 - ☒ 2. ₹19785
 - ☒ 3. ₹20050
 - ☒ 4. ₹19500

Question ID : 441009548026

Chosen Option : 4

Q.21 What is the difference between the average of prime numbers between 40 and 70 and the average of multiples of 5 between 20 and 50?

- Ans
- ☒ 1. 18
 - ☒ 2. 10
 - ☒ 3. 21
 - ☒ 4. 12

Question ID : 441009580941

Chosen Option : 1

Q.22 The monthly incomes of two friends, Salil and Mukund, are in the ratio 3 : 4. Each of them saves ₹24,000 every month. If the ratio of their monthly expenditures is 1 : 4, find the monthly income of Salil (in ₹).

- Ans
- ☒ 1. 28,000
 - ☒ 2. 37,800
 - ☒ 3. 26,000
 - ☒ 4. 27,000

Question ID : 441009854151

Chosen Option : 4

Q.23 Two taps can fill a cistern in 2 hours and 43 hours, respectively. A third tap can empty it in 43 hours. How long (in hours) will it take to fill half of the empty cistern, if all of them are opened together?

- Ans
- ☐ 1. 2
 - ☒ 2. 1
 - ☐ 3. 4
 - ☐ 4. 3

Question ID : 441009518585
Chosen Option : 2

Q.24 Find the simple interest (in ₹) if a sum of ₹500 is borrowed for 7.5 years at 16% per annum rate of interest.

- Ans
- ☒ 1. 600
 - ☐ 2. 650
 - ☐ 3. 700
 - ☐ 4. 580

Question ID : 441009852059
Chosen Option : 1

Q.25 Two ropes of lengths 84 metres and 126 metres are to be cut into pieces of equal length. What is the greatest possible length of each piece?

- Ans
- ☐ 1. 56 metres
 - ☐ 2. 21 metres
 - ☒ 3. 42 metres
 - ☐ 4. 84 metres

Question ID : 4410093721
Chosen Option : 3

Section : Computer knowledge

Q.26 If you need to calculate the average of values in a list where only certain entries meet a specific condition, which Excel function is most appropriate?

- Ans
- ☐ 1. MAX
 - ☐ 2. COUNT
 - ☐ 3. SUM
 - ☒ 4. AVERAGEIF

Question ID : 4410092273012
Chosen Option : 4

Q.27 Which statement best describes a Wide Area Network (WAN)?

- Ans
- ☐ 1. It links devices limited to one room
 - ☐ 2. It is used only for sharing resources in one classroom
 - ☒ 3. It connects computers over large geographical areas
 - ☐ 4. It connects devices only within a single building

Question ID : 4410092272834
Chosen Option : 3

Q.28 Which option should you choose to apply a template only to certain slides and not the whole presentation in MS PowerPoint?

- Ans
- ☐ 1. Use Reset Slide option in Home tab
 - ☐ 2. Click the theme to apply to all slides
 - ☒ 3. Right-click the theme and select Apply to Selected Slides
 - ☐ 4. Apply template from File menu

Question ID : 4410092158547
Chosen Option : 3

Q.29 A user needs to quickly locate a presentation file created last month but cannot recall its name. Which approach most efficiently utilizes Windows OS features to narrow down results within the Documents folder?

- Ans
- ☐ 1. Sort the Documents folder by file size to locate large files
 - ☐ 2. Manually browse each subfolder inside Documents
 - ☒ 3. Apply a date filter in the search box to restrict results to last month's files
 - ☐ 4. Search by file extension in the Documents folder

Question ID : 4410092276309
Chosen Option : 3

Q.30 An architect is updating a project proposal using a word processor. To emphasize specific legal clauses, she chooses to apply a particular formatting style. Which action would most effectively ensure these clauses are consistently highlighted throughout the document, even if the text is later edited or new content is inserted?

- Ans
- ☐ 1. Creating a custom header for each legal clause and updating as needed
 - ☐ 2. Using the format painter tool on each clause once content is finalized
 - ☒ 3. Assigning a predefined style to each clause before making further edits
 - ☐ 4. Manually changing the font color of each clause after editing

Question ID : 4410092276329
Chosen Option : 1

Q.31 In an AC induction motor drive, the relation between the torque, T_e , and the stator supply voltage, V_1 , is _____.

Ans

☐ 1. $T_e \propto \frac{1}{V_1^2}$

☒ 2. $T_e \propto V_1^2$

☐ 3. $T_e \propto V_1$

☐ 4. $T_e \propto \frac{1}{V_1}$

Question ID : 441009111605
Chosen Option : 2

Q.32 A DC potentiometer is used for the measurement of an unknown resistance. During measurement, following readings were observed:

Voltage drop across unknown resistance = 0.6125 V.

Voltage drop across 0.1 Ω standard resistance = 1.225 V.

What is the value of the unknown resistance?

Ans

☐ 1. 0.2 Ω

☐ 2. 0.02 Ω

☐ 3. 0.5 Ω

☒ 4. 0.05 Ω

Question ID : 4410091536506
Chosen Option : 4

Q.33 The working principle of a single-phase transformer is based on:

Ans

☐ 1. Electrostatic induction

☐ 2. Self-induction

☐ 3. Lorentz force law

☒ 4. Mutual induction

Question ID : 4410091557275
Chosen Option : 4

Q.34 What are the two parameters that can be adjusted to balance the unknown voltage in a Drysdale Polar Potentiometer?

Ans

☐ 1. Magnitude and Frequency of the voltage

☐ 2. Resistive and Inductive components of the voltage

☐ 3. Real and Imaginary components of the voltage

☒ 4. Magnitude and Phase angle of the voltage

Question ID : 4410091544790
Chosen Option : 1

Q.35 Which method is commonly used to improve the load-bearing capacity of weak soils under substation foundations?

- Ans
- ☐ 1. Increasing foundation weight
 - ☒ 2. Compaction of the soil layers
 - ☐ 3. Using lighter transmission towers
 - ☐ 4. Widening the pedestal above foundation level

Question ID : 4410091813275
Chosen Option : 4

Q.36 A GTO is used as a controlled switch in a forced-commutated converter. A junior engineer assumes that once the commutation circuit forces the anode current to zero, no further gate action is required. Under which of the following conditions would this assumption lead to unreliable operation?

- Ans
- ☐ 1. When the gate-cathode junction is reverse biased after GTO turn-off
 - ☒ 2. When there is large amount of stored charge present in the GTO after turn-off
 - ☐ 3. When the GTO operates at low switching frequency
 - ☐ 4. When a snubber circuit is connected across the GTO

Question ID : 4410091555229
Chosen Option : 2

Q.37 In an AC clamp-on meter, which component functions as the 'single-turn primary winding'?

- Ans
- ☐ 1. The secondary coil inside the meter's jaws.
 - ☐ 2. The internal circuitry of the meter.
 - ☐ 3. The iron core in the jaws.
 - ☒ 4. The conductor (wire) being clamped around.

Question ID : 4410091544753
Chosen Option : 4

Q.38 What is meant by the statement that "no part of the power cycle at any time becomes negative" in a purely resistive circuit?

- Ans
- ☐ 1. The current is very small.
 - ☐ 2. The circuit has a large inductive or capacitive component.
 - ☐ 3. The circuit only uses DC power.
 - ☒ 4. All the electrical energy supplied by the source is continuously dissipated as heat.

Question ID : 4410091546011
Chosen Option : 4

Q.39 The primary operating principle of a DC potentiometer relies on _____.

- Ans
- ☐ 1. Ohm's Law, measuring the voltage drop across a known resistance.
 - ☐ 2. The electromagnetic induction, generating a counter-EMF to balance the unknown voltage.
 - ☐ 3. Kirchhoff's Current Law, ensuring that the current in the circuit remains constant.
 - ☒ 4. Comparing an unknown EMF with a known reference voltage using a null-deflection method.

Question ID : 4410091544797
Chosen Option : 3

Q.40 Which of the following statements best explains why Kirchhoff's Voltage Law (KVL) holds true in a closed electrical loop?

- Ans
- ☐ 1. Because charge accumulate at any point in an ideal conducting loop
 - ☒ 2. Because electric energy supplied and consumed around a closed path must sum to zero
 - ☐ 3. Because the magnetic flux linking the loop is always constant in all practical circuits
 - ☐ 4. Because the algebraic sum of all electric currents entering a node must be zero

Question ID : 4410091557371
Chosen Option : 2

Q.41 A Hay's bridge used for measuring the inductance (L_1) and resistance (R_1) of a coil is balanced at 500 Hz. If the frequency is shifted to 1000 Hz without changing any bridge components, which of the following is true?

- Ans
- ☒ 1. The bridge becomes unbalanced because the balancing expressions for L_1 and R_1 involve the frequency term.
 - ☐ 2. The balance point remains the same if the quality factor is less than 10.
 - ☐ 3. The bridge remains balanced because the ratio of impedances is constant.
 - ☐ 4. Only the balance point for resistance R_1 is affected by the change in frequency.

Question ID : 4410091533492
Chosen Option : 3

Q.42 For a 1-line to 4-line DEMUX with data input D, select bits S1 (MSB), S0 (LSB) and active-high outputs Y3.... Y0, when S1S0 = 10 and D = 1, the output pattern of Y3 Y2 Y1 Y0 is _____.

- Ans
- ☐ 1. 1000
 - ☐ 2. 0001
 - ☒ 3. 0100
 - ☐ 4. 0010

Question ID : 4410091558269
Chosen Option : 4

Q.43 Which of the following statements describes the output behaviour of a NOT gate when the input signal changes from LOW to HIGH?

- Ans ☒ 1. The output transitions from HIGH to LOW.
- ☐ 2. The output becomes undefined.
- ☐ 3. The output transitions from LOW to HIGH.
- ☐ 4. The output remains unchanged.

Question ID : 4410091792310
Chosen Option : 1

Q.44 In order to maintain constant flux in the motor during stator frequency control, the inverter must _____

- Ans ☒ 1. Vary stator voltage proportionally with frequency
- ☐ 2. Use PWM to eliminate harmonics
- ☐ 3. Increase stator current with frequency
- ☐ 4. Keep stator voltage constant

Question ID : 4410091559996
Chosen Option : 1

Q.45 In Parallel (R-L and C) circuit, what is the value of effective impedance and Q-factor at resonance?

- Ans ☐ 1. $\frac{R}{C}, \frac{1}{L} \sqrt{\frac{1}{RC}}$
- ☐ 2. $R, \frac{1}{R} \sqrt{\frac{L}{C}}$
- ☒ 3. $\frac{L}{CR}, \frac{1}{R} \sqrt{\frac{L}{C}}$
- ☐ 4. $\frac{LR}{C}, \frac{1}{R} \sqrt{\frac{C}{L}}$

Question ID : 4410091546003
Chosen Option : 2

Q.46 At resonance, the impedance of a series RLC circuit is _____.

- Ans ☒ 1. Minimum
- ☐ 2. Equal to capacitive reactance
- ☐ 3. Equal to inductive reactance
- ☐ 4. Maximum

Question ID : 4410091546061
Chosen Option : 1

Q.47 A designer must build a 3-bit asynchronous mod-5 counter using T flip-flops, where each flip-flop toggles when driven by the output of the previous stage. Which design choice ensures correct recycling from the invalid states 101_2 , 110_2 , and 111_2 back to 000_2 ?

- Ans
- ☒ 1. Use combinational decoding of 101_2 to produce an asynchronous clear
 - ☐ 2. Use a toggle-enable gate on the LSB controlled by the decoded state 010_2
 - ☐ 3. Use a synchronous load input triggered by 101_2
 - ☐ 4. Use an asynchronous preset to force the MSB high when the count reaches 011_2

Question ID : 4410091538916

Chosen Option : 4

Q.48 A parallel RLC network is operating slightly below its resonant frequency. Which of the following statements correctly describes its behavior?

- Ans
- ☐ 1. The circuit appears capacitive and its impedance increases as frequency approaches resonant frequency
 - ☐ 2. The circuit appears inductive but its impedance remains constant as frequency approaches resonant frequency
 - ☐ 3. The circuit appears capacitive and its impedance decreases as frequency approaches resonant frequency
 - ☒ 4. The circuit appears inductive and its impedance increases as frequency approaches resonant frequency

Question ID : 4410091565484

Chosen Option : 4

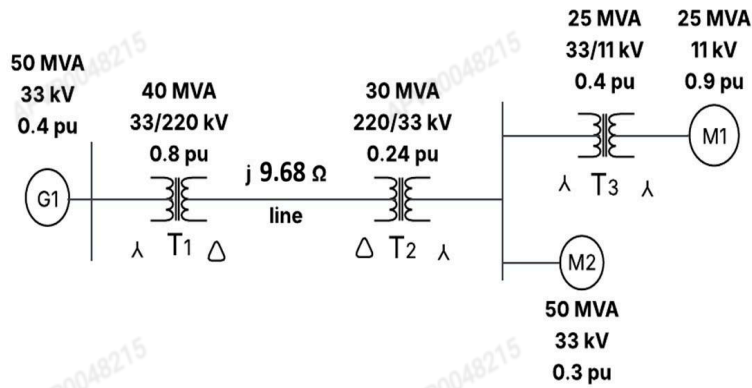
Q.49 In a 25 kV AC traction system, the neutral section is provided primarily to _____.

- Ans
- ☐ 1. Allow regenerative braking without interference
 - ☒ 2. Prevent circulating currents between adjacent substations
 - ☐ 3. Reduce corona losses in overhead lines
 - ☐ 4. Minimize voltage drop during peak load

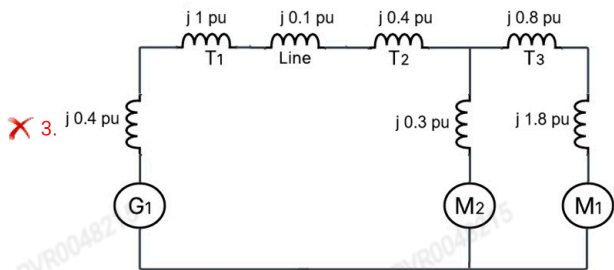
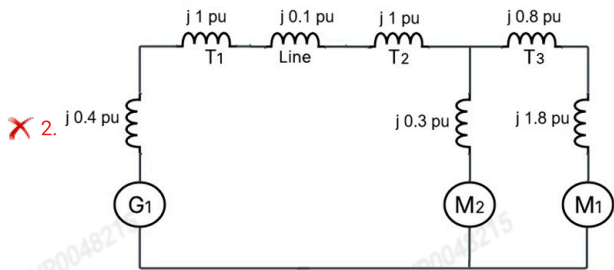
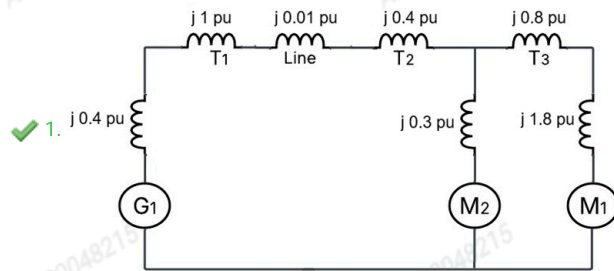
Question ID : 4410091559622

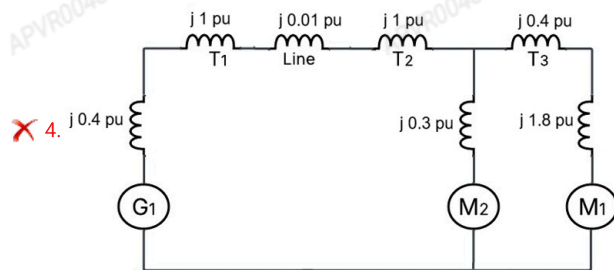
Chosen Option : 1

Q.50 The power system shown in the given figure has the ratings and per unit reactances as given below. All the resistances have been neglected, and the per unit reactance of each component has been calculated on its respective rated values. The transmission line reactance is 9.68Ω . Identify the correct per-unit reactance diagram of the system from the given options, taking a common base of 50 MVA and taking base voltage for the line section as 220 kV (line-to-line) and base voltages for other sections taken accordingly.



Ans





Question ID : 4410091572732
Chosen Option : 1

Q.51 Which of the following construction practices is most effective in minimizing the risk of honeycombing in concrete foundations for substations?

- Ans
- ☒ 1. Ensuring proper vibration during placing
 - ☐ 2. Using larger aggregate
 - ☐ 3. Increasing water-cement ratio
 - ☐ 4. Reducing cement content

Question ID : 4410091816701
Chosen Option : 1

Q.52 If the peak current (I_m) in a half wave rectifier circuit is 10 A, what is the average output current (I)?

- Ans
- ☐ 1. $I = 10 \text{ A}$
 - ☒ 2. $I = 10 / \pi \text{ A}$
 - ☐ 3. $I = 5 \text{ A}$
 - ☐ 4. $I = 20 / \pi \text{ A}$

Question ID : 4410091809811
Chosen Option : 2

Q.53 Which of the following is a key advantage of indirect resistance heating over direct resistance heating?

- Ans
- ☐ 1. Elimination of the need for insulation
 - ☐ 2. Higher efficiency due to direct current flow
 - ☐ 3. Reduced thermal losses in the heating element
 - ☒ 4. Ability to heat non-conducting materials uniformly

Question ID : 4410091555735
Chosen Option : 4

Q.54 Choose the correct statements for the peak detector circuit.

S1: A basic peak detector circuit usually consists of diode and capacitor.
S2: The output of an ideal peak detector is the maximum value of the input.
S3: The time constant of a peak detector is only C.

- Ans ☐ 1. Only S1 and S3.
- ☒ 2. Only S1 and S2.
- ☐ 3. S1,S2 and S3
- ☐ 4. Only S3 and S2.

Question ID : 4410091562491

Chosen Option : 3

Q.55 What would happen to the armature current of a DC motor if the back EMF is suddenly lost or drops drastically (becomes close to zero) while the supply voltage remains constant?

- Ans ☐ 1. It would decrease to a much lower value.
- ☒ 2. It would become dangerously high.
- ☐ 3. It would keep fluctuating slightly in value for some time before settling on a lower value.
- ☐ 4. It would remain at nearly the same value as before.

Question ID : 4410091545934

Chosen Option : 2

Q.56 If surface water naturally drains toward a proposed construction site for a foundation, which site preparation measure is most effective to prevent water-related problems?

- Ans ☐ 1. Reduce foundation dimensions to minimize water exposure
- ☒ 2. Provide drains or grade the ground to divert water away
- ☐ 3. Increase foundation depth and use coarse fill at the base
- ☐ 4. Add sand or gravel layers to improve surface drainage without grading

Question ID : 4410091813259

Chosen Option : 4

Q.57 The self GMD (Geometric Mean Distance) of a conductor used in transmission line inductance calculation is equivalent to _____.

- Ans ☐ 1. The spacing distance value measured between the two conductors in the line
- ☐ 2. The arithmetic mean radius value calculated from conductor diameters
- ☒ 3. The geometric mean radius value defined for the conductor geometry
- ☐ 4. The physical radius value considered for the conductor geometry

Question ID : 4410091563932

Chosen Option : 3

Q.58 In the transient response of a second order system, delay time t_d is the time required for the response to reach the _____ of the final value the very first time.

- Ans
- ☒ 1. 10%
 - ☒ 2. 50%
 - ☒ 3. 95%
 - ☒ 4. 90%

Question ID : 4410091661140
Chosen Option : 2

Q.59 A 10-bit DAC has a full-scale output voltage of 10.23 V. What is its resolution in mV?

- Ans
- ☒ 1. 20 mV
 - ☒ 2. 100 mV
 - ☒ 3. 10 mV
 - ☒ 4. 1 mV

Question ID : 4410091562912
Chosen Option : 2

Q.60 In a three-phase dual converter drive, the positive converter determines the motor's operation in which of the following modes?

- Ans
- ☒ 1. Zero-torque holding mode
 - ☒ 2. Forward rotation of the motor
 - ☒ 3. Regenerative braking in reverse direction
 - ☒ 4. Reverse motoring mode

Question ID : 4410091555223
Chosen Option : 3

Q.61 Which property of building bricks makes them suitable for accommodating electrical conduits?

- Ans
- ☒ 1. Electrical conductivity for grounding purposes
 - ☒ 2. High thermal conductivity for heat dissipation
 - ☒ 3. Uniform shape and adequate compressive strength
 - ☒ 4. High porosity for better ventilation

Question ID : 4410091787031
Chosen Option : 2

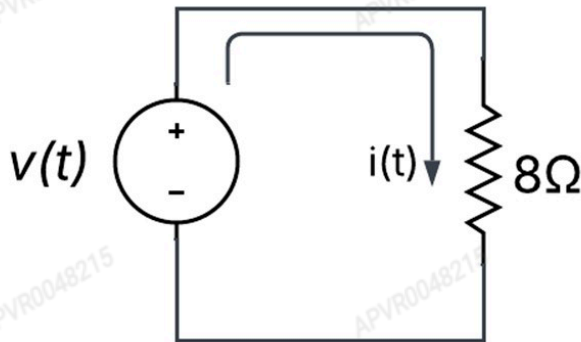
Q.62 Two identical single-phase induction motors operate at the same speed and load, but one draws higher input power under no-load conditions. From the standpoint of equivalent circuit modeling, which of the following factors can be the cause for this difference?

- Ans
- ☒ 1. Greater rotor copper losses due to increased slip
 - ☒ 2. Increased mechanical friction due to bearing wear
 - ☒ 3. Larger backward field current due to phase imbalance
 - ☒ 4. Higher magnetic losses in the stator due to core properties

Question ID : 4410091549472

Chosen Option : 2

Q.63 Consider the circuit shown, where the input voltage is $v(t) = 4 u(t-3)$ V and the resistor value is $R = 8 \Omega$. What will be the expression for the current $i(s)$ in the s-domain (obtained using Laplace Transform)?



- Ans
- ☒ 1. $\frac{1}{2s} e^{-3s}$
 - ☒ 2. $\frac{1}{2s} e^{-s}$
 - ☒ 3. $8s e^{-3s}$
 - ☒ 4. $\frac{4}{s}$

Question ID : 4410091565110

Chosen Option : 1

Q.64 Consider a 1-to-4 demultiplexer where the data input is permanently tied to logic '1'. If the selection lines are driven by two binary variables, X and Y, which of the following standard combinational logic functions does this circuit strictly perform?

- Ans
- ☒ 1. 4-to-2 Encoder
 - ☒ 2. 2-bit Magnitude Comparator
 - ☒ 3. 2-to-4 Decoder
 - ☒ 4. 4-to-1 Multiplexer

Question ID : 4410091538805

Chosen Option : 2

Q.65 _____ is primarily impacted by hysteresis loss.

Ans

- ☒ 1. Mechanical efficiency
- ☒ 2. Magnetic parts of electrical machines
- ☒ 3. Voltage stability
- ☒ 4. Cooling system

Question ID : 4410091545439
Chosen Option : 2

Q.66 In an asynchronous BCD ripple counter consisting of four flip-flops, if the propagation delay of a single flip-flop is $t_{pd} = 20$ ns, what is the theoretical maximum input clock frequency f_{max} allowed to ensure the counter reads the transition from State 9 (1001) to State 0 (0000) correctly without decoding errors?

Ans

- ☒ 1. 25 MHz
- ☒ 2. 50 MHz
- ☒ 3. 100 MHz
- ☒ 4. 12.5 MHz

Question ID : 4410091538909
Chosen Option : 4

Q.67 For a second-order system with natural frequency ω_n and damping ratio ζ , what is the effect on the resonant frequency ω_r if ω_n is held constant while ζ is increased?

Ans

- ☒ 1. ω_r decreases
- ☒ 2. ω_r becomes imaginary
- ☒ 3. ω_r remains unchanged
- ☒ 4. ω_r increases

Question ID : 4410091553411
Chosen Option : 3

Q.68 Campbell's modification of Heaviside bridge is used for accurate measurement of:

Ans

- ☒ 1. Self-inductance in terms of mutual inductance
- ☒ 2. Capacitance
- ☒ 3. Dielectric strength
- ☒ 4. Resistance

Question ID : 4410091547311
Chosen Option : 1

Q.69 A civil engineer must select a concrete mix for a substation foundation in a cold region. Which of the following measures should be prioritized to ensure long-term durability?

- Ans
- ☐ 1. Reduce mixing water to the lowest possible level
 - ☐ 2. Add steel fibers to enhance crack-bridging flexibility
 - ☐ 3. Increase cement content to maximize paste strength only
 - ☒ 4. Use air-entrained concrete to improve freeze-thaw resistance

Question ID : 4410091790943
Chosen Option : 4

Q.70 During a Gauss-Seidel iteration, if a PV (or voltage-controlled) bus's reactive power surpasses certain limits, the bus is transformed into _____.

- Ans
- ☐ 1. Slack bus
 - ☐ 2. Transmission bus
 - ☒ 3. PQ (or load) bus
 - ☐ 4. Swing bus

Question ID : 4410091547042
Chosen Option : 3

Q.71 When excitation of an alternator operating in parallel is increased, its:

- Ans
- ☐ 1. Real power increases
 - ☐ 2. Frequency increases
 - ☐ 3. Speed increases
 - ☒ 4. Reactive Power Increases

Question ID : 4410091555053
Chosen Option : 4

Q.72 In 120° mode VSI, how many switches conduct at a time?

- Ans
- ☒ 1. Two
 - ☐ 2. Three
 - ☐ 3. One
 - ☐ 4. All six

Question ID : 4410091564033
Chosen Option : 1

Q.73 What is the primary function of a SCADA system in industrial automation?

- Ans
- ☐ 1. To manufacture sensors
 - ☐ 2. To increase energy consumption
 - ☐ 3. To replace human operators entirely
 - ☒ 4. To monitor and control industrial processes

Question ID : 4410091805400
Chosen Option : 4

Q.74 "Any set of n-phase unbalanced system can be resolved into (n-1) balanced systems and 1 zero sequence system". Which theorem has this statement?

Ans

- ☒ 1. Fortesque's theorem
- ☐ 2. Voltage balancing theorem
- ☐ 3. Sequence analyzing theorem
- ☐ 4. Forte's theorem

Question ID : 4410091560727
Chosen Option : 1

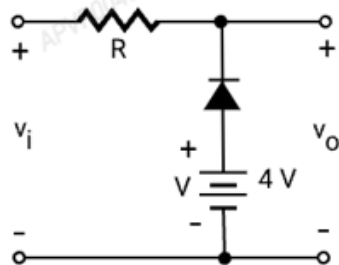
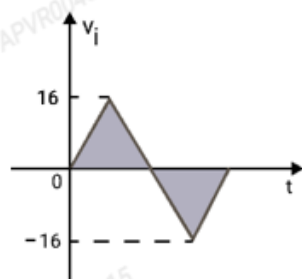
Q.75 A double cage rotor in a three-phase induction motor is designed to improve starting torque and running efficiency. What is the relative resistance of the two cages?

Ans

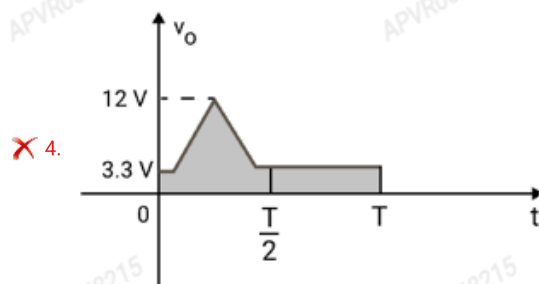
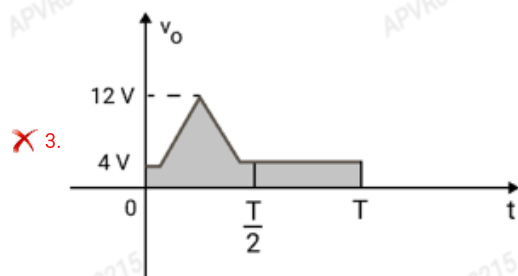
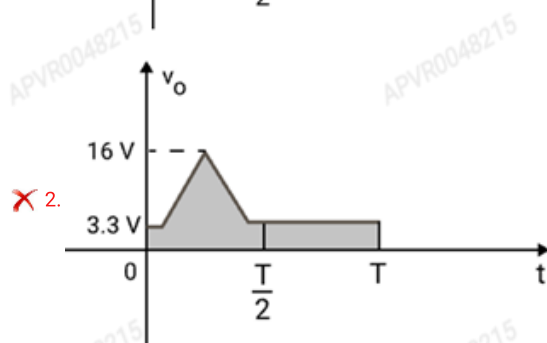
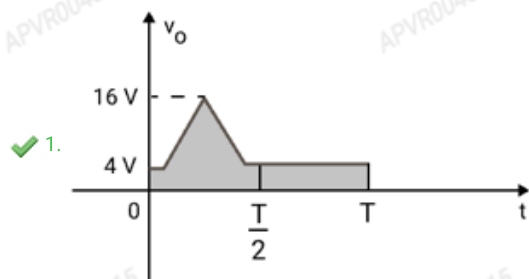
- ☐ 1. the inner cage has high resistance and outer cage has low resistance.
- ☐ 2. both the cages have low resistances.
- ☐ 3. both the cages have high resistances.
- ☒ 4. the inner cage has low resistance and outer cage has high resistance.

Question ID : 4410091545694
Chosen Option : 4

Q.76 Assuming the diode to be ideal, choose the correct output waveform.



Ans



Question ID : 4410091567081
Chosen Option : 1

Q.77 The critical clearing angle refers to:

- Ans
- ☐ 1. Angle at which transmission line trips
 - ☐ 2. Generator excitation angle
 - ☐ 3. Minimum load angle
 - ☒ 4. Maximum rotor angle permissible before losing synchronism

Question ID : 4410091561273
Chosen Option : 4

Q.78 Which among the following is a commonly used industrial IGBT structure?

- Ans
- ☒ 1. Punch-Through IGBT
 - ☐ 2. Schottky IGBT
 - ☐ 3. BJT
 - ☐ 4. MCT

Question ID : 4410091563097
Chosen Option : 1

Q.79 If the Nyquist plot does not encircle the point $(-1, 0)$, the system is _____.

- Ans
- ☐ 1. Stable only if it crosses real axis
 - ☒ 2. Stable if open-loop has no Right Half Plane (RHP) poles
 - ☐ 3. Always unstable
 - ☐ 4. Always marginally stable

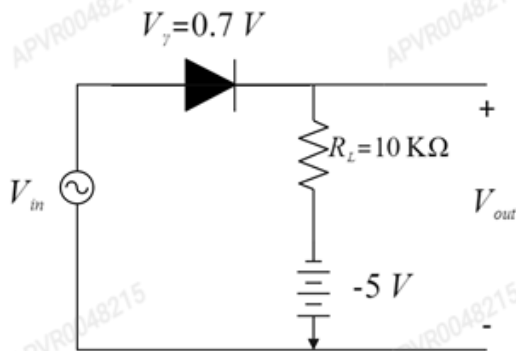
Question ID : 4410091554072
Chosen Option : 2

Q.80 Which of the following is the primary effect of source impedance in a single-phase full converter?

- Ans
- ☒ 1. Reduction in average output voltage
 - ☐ 2. Increase in average output voltage
 - ☐ 3. No change in output voltage
 - ☐ 4. Increase in harmonic distortion only

Question ID : 4410091550393
Chosen Option : 1

Q.81 Consider a series clipper designed to pass signals above a reference level V_{ref} and clip everything below it. A silicon diode having a cut-in voltage of 0.7 V has its anode connected to the input V_{in} and its cathode connected to the output, as shown in the figure. A load resistor R_L connects the output node to a voltage of -5 V . The input is a sinusoid with a peak amplitude of 10 V . What is the voltage level of the "flat" portion of the output waveform?



Ans ☒ 1. -5.0 V

☒ 2. -5.7 V

☒ 3. -4.3 V

☒ 4. 0 V

Question ID : 4410091528664
Chosen Option : 2

Q.82 A good-quality electric weld should have:

Ans ☒ 1. Uniform strength and free from cracks

☒ 2. High internal stresses

☒ 3. Poor fusion at joint surfaces

☒ 4. Brittle structure and porosity

Question ID : 4410091562228
Chosen Option : 1

Q.83 The frequency response of a linear system $G(j\omega)$ is provided in the tubular form below.

$G(j\omega)$	1.3	1.2	1.0	0.8	0.5	0.3
$\angle G(j\omega)$	-130°	-140°	-150°	-160°	-180°	-200°

The approximate gain margin and phase margin of the system are _____.

Assume following log values with base 10:

$\log 2 = 0.3010$; $\log 3 = 0.4771$; $\log 4 = 0.6020$.

- Ans
- ☐ 1. -6 db and -30°
 - ☐ 2. 6 db and -30°
 - ☒ 3. 6 db and 30°
 - ☐ 4. -6 db and 30°

Question ID : 4410091661806
Chosen Option : 3

Q.84 Increasing the number of pulses in a PWM strategy reduces:

- Ans
- ☒ 1. Harmonic distortion
 - ☐ 2. Switching frequency
 - ☐ 3. Fundamental component
 - ☐ 4. Output voltage

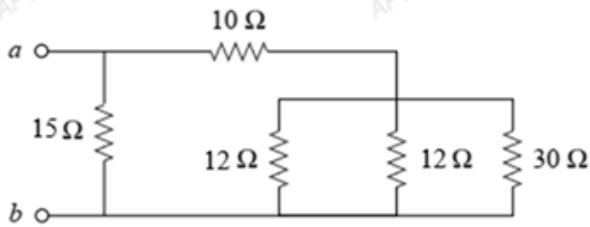
Question ID : 4410091563134
Chosen Option : 2

Q.85 In a digital system, which logic function will result when the outputs of two NOR gates are fed as inputs to a third NOR gate? (Assume each of the first two NOR gates is used as an inverter)

- Ans
- ☐ 1. The logic function is equivalent to an OR gate.
 - ☒ 2. The logic function is equivalent to an AND gate.
 - ☐ 3. The logic function is equivalent to a NAND gate.
 - ☐ 4. The logic function is equivalent to an XOR gate.

Question ID : 4410091792301
Chosen Option : 2

Q.86 For the circuit given below, find the equivalent resistance between the terminals 'ab'.



- Ans
- ☐ 1. 22.5 Ω
 - ☒ 2. 7.5 Ω
 - ☐ 3. 30 Ω
 - ☐ 4. 15 Ω

Question ID : 4410091554848
Chosen Option : 2

Q.87 High torque-to-weight ratio in PMMC meters results in:

- Ans
- ☒ 1. Higher deflecting torque for a given current, improving scale sensitivity and accuracy
 - ☐ 2. More rapid and steady pointer settling due to effective electromagnetic damping
 - ☐ 3. Increased ability of the moving coil to withstand momentary overload currents
 - ☐ 4. Increased ability of the instrument to measure alternating current without additional circuitry

Question ID : 4410091545855
Chosen Option : 1

Q.88 Which feature most directly allows electro-dynamometer instruments to remain free from hysteresis and eddy current errors?

- Ans
- ☐ 1. High-permeability laminated iron cores
 - ☐ 2. Permanent magnet flux system
 - ☐ 3. Aluminium damping drum around the coil
 - ☒ 4. Air-cored fixed and moving coils

Question ID : 4410091537599
Chosen Option : 4

Q.89 What is the main advantage of feeding a distributor from both ends compared to only using one end?

- Ans
- ☒ 1. A significantly reduced voltage drops.
 - ☐ 2. It decreases load-carrying capacity.
 - ☐ 3. A simpler protection system.
 - ☐ 4. The distributor is guaranteed to never fail.

Question ID : 4410091547491
Chosen Option : 1

Q.90 Choose the correct statement for the effect of derivative-D controller in the control system.

S1: Steady state error increases by the derivative control action.

S2: The effective damping is increased in the D controller.

S3: The maximum overshoot reduces.

Ans ☒ 1. Only S2 and S3.

☐ 2. S1,S2 and S3

☐ 3. Only S2 and S1.

☐ 4. Only S1 and S3.

Question ID : 4410091563378

Chosen Option : 3

Q.91 Select the option that is true regarding the following two statements about low-frequency three-phase traction supply, labelled Assertion (A) and Reason (R).

Assertion (A): A transformer is not required in the locomotive in a three-phase low-frequency supply system.

Reason (R): The line voltage used in this system is already suitable for feeding the induction motor.

Ans ☐ 1. Statements A and R are both true, but R is not the correct explanation of A.

☒ 2. Statements A and R are both true, and R is the correct explanation of A.

☐ 3. Statement A is false, but statement R is true.

☐ 4. Statement A is true, but statement R is false.

Question ID : 4410091555201

Chosen Option : 1

Q.92 Voltage-shunt feedback results in _____.

Ans ☐ 1. No effect

☒ 2. Decreased input impedance

☐ 3. Increased input impedance

☐ 4. Increased output impedance

Question ID : 4410091553576

Chosen Option : 2

Q.93 For the transfer function $G(s) = \frac{100}{(s + 10)}$, the asymptotic magnitude slope for $\omega \gg 10$ is_____

Ans ☒ 1. -20 dB/decade

☐ 2. 0 dB/decade

☐ 3. -40 dB/decade

☐ 4. +20 dB/decade

Question ID : 4410091576392

Chosen Option : 1

Q.94 The capacitance between two conductors in a 3-core cable is measured by _____.

- Ans
- ☒ 1. Measuring each conductor separately with respect to sheath
 - ☒ 2. Measuring all three conductors together against the sheath
 - ☒ 3. Connecting the 3rd conductor and sheath together to form a common return
 - ☒ 4. Using only the sheath as one terminal and conductor as the other

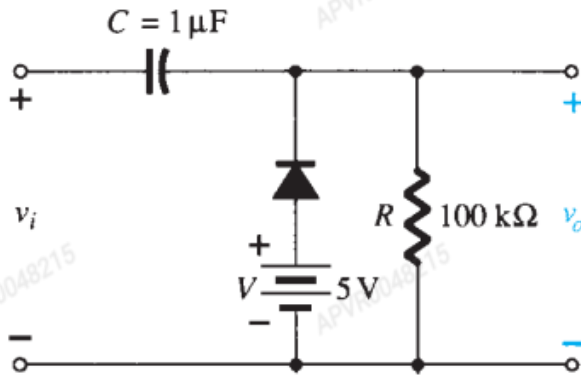
Question ID : 4410091565336
Chosen Option : 3

Q.95 The base values typically selected while calculating per-unit reactance (or any per-unit quantity) for a power system are _____.

- Ans
- ☒ 1. kVA and kV
 - ☒ 2. kW and ohm
 - ☒ 3. Resistance and quality factor
 - ☒ 4. Current and power factor

Question ID : 4410091547044
Chosen Option : 1

Q.96 In a clamper circuit, a battery of voltage 5V is placed in series with the diode as shown in the figure. Which of the following statements correctly describes how this battery modifies the clamping action?



- Ans
- ☒ 1. The output is shifted upward by exactly 5V.
 - ☒ 2. The output clamps at a level determined by the input's average value, independent of the battery.
 - ☒ 3. The output is shifted downward by exactly 5V.
 - ☒ 4. The battery reduces the capacitor's charging voltage, causing the output to clamp at a value lower than the original zero reference.

Question ID : 4410091528653
Chosen Option : 4

Q.97 In a DC motor, if the current in the armature is reversed along with the direction of the magnetic field, then according to the Fleming's Left-Hand Rule _____.

- Ans
- ☒ 1. The direction of rotation will reverse
 - ☒ 2. The direction of rotation will be same
 - ☒ 3. The motor torque will be doubled
 - ☒ 4. The motor will stop rotating

Question ID : 4410091563505
Chosen Option : 2

Q.98 A 4-pole, 50 Hz single-phase induction motor runs at 1000 rpm. What is the value of backward slip of the motor?

- Ans
- ☒ 1. 5/3
 - ☒ 2. 1/3
 - ☒ 3. 4/3
 - ☒ 4. 2/3

Question ID : 4410091563544
Chosen Option : 1

Q.99 A half-wave rectifier without a filter is supplied with a 60 Hz AC input. What is the frequency of the rectified output waveform?

- Ans
- ☒ 1. 30 Hz
 - ☒ 2. 120 Hz
 - ☒ 3. 60 Hz
 - ☒ 4. 0 Hz

Question ID : 4410091805394
Chosen Option : 3

Q.100 The power factor of an AC circuit is defined as the:

- Ans
- ☒ 1. Ratio of real power to apparent power
 - ☒ 2. Ratio of reactive power to apparent power
 - ☒ 3. Ratio of real power to reactive power
 - ☒ 4. Ratio of voltage to current

Question ID : 4410091557342
Chosen Option : 1